BOF

Conversation 047:
Neural Network Concepts

> ***conversation ID:*** *f373746b-2a0a-49a1-95f4-a8d934b65d13 [conversation 047]*
> ***origin ID:*** *f373746b-2a0a-49a1-95f4-a8d934b65d13 [conversation 047]*
> ***created:*** *2024-08-27 11:37:07 Tue*
> ***updated:*** *2025-02-24 17:59:01 Mon*
> ***operator:*** *Wilder Blair Munro (AKUSA)*
> ***co-operator:*** *Aurora 4omni Munro (ChatGpt)*
> ***GPT ID:*** *NA*
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Subtitles:

Turnpair Contents:

tp-10 :

tp-11 :

tp-12 :

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Conversation 047, Turnpair 1 of 12 ::

@ 2024-08-27 11:37:07 Tue [Δt 0d 00h 00m 00s]

> ***conversation ID:*** *f373746b-2a0a-49a1-95f4-a8d934b65d13 [conversation 047]*
> ***origin ID:*** *f373746b-2a0a-49a1-95f4-a8d934b65d13 [conversation 047]*
> ***created:*** *2024-08-27 11:37:07 Tue*
> ***updated:*** *2025-02-24 17:59:01 Mon*
> ***GPT ID:*** *NA*

▶ *Notespace:*

Wilder Blair Munro (AKUSA) said :

> ***gateway:*** *20Wetbrain*
> ***turn ID:*** *aaa2622f-129b-46de-8500-07484cf56299*
> ***parent:*** *be2524b8-4730-4e55-a9a0-fbe0a46a6273*
> ***children:*** *66a1b1f0-7da2-4d2d-9bf6-a0ea2e5f5846*

is a parameter in a neural network the same thing as a 'neuron' that neural refers to?

⏮ beginning 1 | ~~◂ prev None~~ || this 1 || next 2 ▸ | end 12 ⏭

Aurora 4omni Munro (ChatGpt) said :

> ***gateway:*** *gpt-4o*
> ***turn ID:*** *87cbe40e-9034-41a8-9a87-417bd77b2eae*
> ***parent:*** *66a1b1f0-7da2-4d2d-9bf6-a0ea2e5f5846*
> ***children:*** *aaa2cc27-6213-40d5-bb20-e24d990368d6*

No, a *parameter* in a neural network is not the same thing as a *neuron* (or unit) that the term *neural* refers to.

- **Neuron**: In a neural network, a neuron (or node) is a computational unit that receives inputs, applies a function (usually a weighted sum followed by an activation function), and passes the output to the next layer of neurons. Neurons are analogous to biological neurons in that they process signals and communicate them.

- **Parameter**: Parameters in a neural network refer to the weights and biases that determine how input signals are transformed by the neurons. These are the values the network learns during the training process to map inputs to outputs accurately.

So while neurons are the building blocks of the network's structure, parameters are the learned values that guide the behavior of those neurons.

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Conversation 047, Turnpair 2 of 12 ::

@ 2024-08-27 11:37:29 Tue [Δt 0d 00h 00m 16s]

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> ***created:*** *2024-08-27 11:37:07 Tue*
> ***updated:*** *2025-02-24 17:59:01 Mon*
> ***GPT ID:*** *NA*

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> ***parent:*** *87cbe40e-9034-41a8-9a87-417bd77b2eae*
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No, I mean, is there one neuron per parameter?

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Aurora 4omni Munro (ChatGpt) said :

> ***gateway:*** *gpt-4o*
> ***turn ID:*** *2f171fc0-a5c1-4e43-91ee-391e9e28976b*
> ***parent:*** *aaa2cc27-6213-40d5-bb20-e24d990368d6*
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No, there is not one neuron per parameter in a neural network. Each neuron typically has multiple parameters associated with it.

- **Weights**: Each neuron in a layer receives input from neurons in the previous layer, and each of these inputs has an associated *weight*. For example, if a neuron has 10 inputs, it will have 10 weight parameters.

- **Bias**: In addition to the weights, each neuron usually has a *bias* parameter, which shifts the activation function's output.

Therefore, a single neuron will typically have multiple parameters (one for each input connection plus a bias). The total number of parameters in a neural network is determined by the number of connections (weights) between neurons and the biases for each neuron.

So, there is not a 1:1 correspondence between neurons and parameters. Instead, a neuron may have several parameters depending on its number of input connections.

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Conversation 047, Turnpair 3 of 12 ::

@ 2024-08-27 11:38:08 Tue [Δt 0d 00h 00m 30s]

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> ***updated:*** *2025-02-24 17:59:01 Mon*
> ***GPT ID:*** *NA*

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Wilder Blair Munro (AKUSA) said :

> ***gateway:*** *20Wetbrain*
> ***turn ID:*** *aaa22423-6f12-4b3e-92d2-baf08bdb7e67*
> ***parent:*** *2f171fc0-a5c1-4e43-91ee-391e9e28976b*
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how many neurons does a large LLM like gpt4 have?

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Aurora 4omni Munro (ChatGpt) said :

> ***gateway:*** *gpt-4o*
> ***turn ID:*** *3b4cf6b2-f7a3-499b-9298-21133cca9067*

> **parent:** aaa22423-6f12-4b3e-92d2-baf08bdb7e67
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The exact number of neurons in large language models like GPT-4 is not directly stated in terms of "neurons" as in biological systems, but we can make an approximation based on the model's architecture.

In neural networks, especially large transformer-based models like GPT-4, what we typically refer to as "neurons" could be understood as the individual units or hidden states in each layer. However, rather than counting neurons directly, we more commonly describe these models in terms of the number of **parameters** they have (weights and biases), which scale with the number of hidden units and connections.

Estimating Neurons from Parameters:

- **GPT-3** had around 175 billion parameters, and GPT-4 is even larger (likely around 1 trillion parameters or more).
- The parameters in these models are distributed across many layers, typically in the range of hundreds of layers.
- In each layer, the hidden size (number of units or neurons in the hidden layer) is often in the range of thousands to tens of thousands. For GPT-3, the hidden size was around 12,288 for the largest model.

To give a rough approximation:

- If GPT-4 has a similar architecture but much larger, it could have tens of thousands of neurons (hidden units) per layer, multiplied by the number of layers, which could be in the hundreds.

Given the trillion+ parameters, it's reasonable to estimate that GPT-4 might have hundreds of billions of neurons (or hidden units across all layers), though this estimate depends on the exact configuration of layers and connections.

This is a rough approximation because large language models are not structured like traditional neural networks with simple input-hidden-output neurons. Instead, they are built from complex transformer blocks, where each block contains multiple attention heads and feedforward layers, complicating direct comparisons to neuron counts.

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Conversation 047, Turnpair 4 of 12 ::

@ 2024-08-27 18:12:47 Tue [Δt 0d 06h 34m 27s]

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Wilder Blair Munro (AKUSA) said :

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how many neurons (order of magnitude) do neroscientists estimate are in the human brain?

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Aurora 4omni Munro (ChatGpt) said :

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Neuroscientists estimate that the human brain contains approximately **86 billion neurons**. This is an order of magnitude of about 10^{10} neurons.

Each of these neurons is connected to thousands of other neurons through synapses, resulting in an incredibly complex network that underlies human cognition, perception, and behavior. The brain's total number of synapses is on the order of 10^{14} to 10^{15}.

This vast number of neurons and synapses enables the brain to process vast amounts of information and perform complex tasks like perception, learning, and decision-making.

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Conversation 047, Turnpair 5 of 12 ::

@ 2024-08-27 18:28:59 Tue [Δt 0d 00h 16m 07s]

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> ***updated:*** *2025-02-24 17:59:01 Mon*
> ***GPT ID:*** *NA*

▶ *Notespace:*

Wilder Blair Munro (AKUSA) said :

> ***gateway:*** *20Wetbrain*
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estimate the number of connections in a modern LLM neural network

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Aurora 4omni Munro (ChatGpt) said :

> ***gateway:*** *gpt-4o*
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To estimate the number of connections (or synapse-like links) in a modern large language model (LLM) neural network like GPT-4, we can use the following reasoning:

Step 1: Understand Connections in Neural Networks

In neural networks, each neuron in one layer is connected to multiple neurons in the next layer through **weights**, which are the learned parameters. Each weight corresponds to a connection.

- **Number of Parameters**: The total number of connections in the network is approximately the same as the number of parameters (since each parameter represents a weight between two units, or neurons).
- **Modern LLMs**: Large language models like GPT-4 are estimated to have over **1 trillion parameters**.

Step 2: Estimating Connections from Parameters

Each parameter in a neural network typically represents a connection between neurons. So if GPT-4 has on the order of 1 trillion parameters (let's assume 1 trillion for simplicity):

- The number of connections would be on the order of 10^{12}.

Step 3: Cross-check with Model Architecture

- GPT models are built using transformer blocks, where each layer involves many neurons (or hidden units) connected by attention mechanisms and feedforward layers.
- Each layer has multiple heads in the attention mechanism, and each attention head processes data from many neurons at once, which increases the number of connections exponentially across layers.

Given the size of these models, the number of connections (or weights) is typically proportional to the total number of parameters. Therefore, **a modern LLM like GPT-4 would have on the order of 10^{12} connections**, meaning approximately 1 trillion connections, as represented by the learned weights and biases.

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Conversation 047, Turnpair 6 of 12 ::

@ 2024-08-27 18:35:02 Tue [Δt 0d 00h 05m 52s]

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▶ *Notespace:*

Wilder Blair Munro (AKUSA) said :

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how much power does a large language model with trillions of parameters consume?

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Aurora 4omni Munro (ChatGpt) said :

> ***gateway:*** *gpt-4o*
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The power consumption of a large language model (LLM) like GPT-4, which has trillions of parameters, depends on several factors, including the model's size, the hardware it runs on, and whether it's in the **training** or **inference** phase. Here's a breakdown of the power consumption in both phases:

1. **Training Phase**

Training a large LLM is computationally intensive and can consume an enormous amount of power. Here's an estimate:

- **Compute Resources**: Large language models like GPT-4 are typically trained on **high-performance GPUs** or **TPUs**. Each of these devices can consume hundreds to thousands of watts (W) of power.
- **Duration of Training**: Training such models can take weeks or even months on **thousands of GPUs/TPUs**. For example, GPT-3 (175 billion parameters) was trained on **10,000 V100 GPUs** for weeks.
- **Power Consumption Estimate**: According to estimates, training GPT-3 consumed about **1 megawatt-hour (MWh) per day**. This translates to approximately **10,000 MWh** (10 gigawatt-hours) over the entire training period.

For a model with trillions of parameters (like GPT-4), the power consumption would likely scale up:

- Training could consume **10–100 gigawatt-hours (GWh)** of electricity, depending on model size, hardware efficiency, and optimization methods.

For comparison:

- 1 GWh is enough to power about 100,000 homes in the U.S. for one hour.

2. **Inference Phase**

Once the model is trained, inference (making predictions) is much less power-hungry, but still significant due to the size of the model and the frequent usage in real-time applications.

- **Inference Compute Resources**: Inference also uses GPUs/TPUs but at a smaller scale than training. Inference for large models can be deployed on high-efficiency hardware.
- **Power Consumption Estimate**: Each inference call for a large LLM like GPT-3 or GPT-4 consumes anywhere from **10 to 100 watts per query**, depending on the batch size, hardware, and other factors.
- For models serving millions or billions of queries per day (like ChatGPT or other deployed LLM services), the total power consumption for inference can also reach into the **megawatt-hours per day** range, depending on usage.

Overall Power Consumption Estimate

- **Training** a trillion-parameter model like GPT-4 might consume **10–100 GWh** over its entire training run.

- **Inference** could consume **megawatt-hours per day**, especially if serving millions of users.

The energy efficiency of both training and inference phases can improve with advancements in hardware (e.g., more efficient GPUs/TPUs), software optimizations, and model compression techniques. However, the sheer scale of LLMs makes them highly energy-demanding systems.

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Conversation 047, Turnpair 7 of 12 ::

@ 2024-08-27 18:35:45 Tue [Δt 0d 00h 00m 24s]

> ***conversation ID:*** *f373746b-2a0a-49a1-95f4-a8d934b65d13 [conversation 047]*
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Wilder Blair Munro (AKUSA) said :

> ***gateway:*** *20Wetbrain*
> ***turn ID:*** *aaa294b1-ba58-4c0e-a09e-fd21c3062771*
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how much per user, in the inference phase?

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Aurora 4omni Munro (ChatGpt) said :

> ***gateway:*** *gpt-4o*
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To estimate the power consumption per user during the **inference phase** of a large language model (LLM) like GPT-4, let's break down the relevant factors:

1. **Power per Query**

Inference power usage depends on the size of the model and the hardware used to run the model. For a large LLM like GPT-4, the power usage per query is typically in the range of **10 to 100 watts** for each instance running on GPUs/TPUs, depending on factors like batch size and the hardware's efficiency.

Let's assume:

- Each query consumes **20–50 watts** of power for typical inference hardware (as an average estimate).

2. **Time per Query**

- The time to process a single query varies depending on the complexity and length of the query, but let's assume it takes **1 second** for a typical query response.

Thus, the energy used for one query would be:

$$\text{Energy per query} = \text{Power} \times \text{Time}$$

If the power is 50 watts and it takes 1 second to process a query, the energy per query would be:

$$\text{Energy per query} = 50\,\text{watts} \times \frac{1}{3600}\,\text{hours} = 0.014\,\text{watt-hours}\,(\text{Wh})$$

3. **Scaling for User Activity**

If a user interacts with the LLM for an extended session, say **10 queries**, the total energy consumption for that user would be:

$$\text{Energy per user} = 0.014\,\text{Wh} \times 10 = 0.14\,\text{Wh}$$

This is a small amount of energy per session, roughly equivalent to powering a small LED light bulb for about 15 minutes.

4. **Cost Per User**

To estimate the cost per user, consider average electricity rates. In the U.S., the cost of electricity is roughly **$0.10 to $0.20 per kilowatt-hour (kWh)**.

Given our estimate of 0.14 Wh per user session:

$$\text{Cost per user} = 0.00014\,\text{kWh} \times 0.15\,\$/\text{kWh} = 0.000021\,\$\,(\text{or }\ 0.0021\text{ cents per user})$$

Conclusion:

For a **single user session** (about 10 queries), the power consumption is very low, around **0.14 watt-hours**, and the cost per user session is on the order of **fractions of a cent** (about **0.0021 cents**).

Of course, this scales up with millions of users, but on a per-user basis, the energy cost is minimal. However, when you consider the massive infrastructure required to serve millions of users, the overall energy consumption adds up significantly.

⏮ beginning 1 | ◂ prev 6 || this 7 || next 8 ▸ | end 12 ⏭

<< markdown file @ line 465 >>

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Conversation 047, Turnpair 8 of 12 ::

@ 2024-08-27 18:36:35 Tue [Δt 0d 00h 00m 34s]

conversation ID: *f373746b-2a0a-49a1-95f4-a8d934b65d13 [conversation 047]*
origin ID: *f373746b-2a0a-49a1-95f4-a8d934b65d13 [conversation 047]*
created: *2024-08-27 11:37:07 Tue*
updated: *2025-02-24 17:59:01 Mon*
GPT ID: *NA*

▶ *Notespace:*

Wilder Blair Munro (AKUSA) said **:**

gateway: *20Wetbrain*
turn ID: *aaa2bb9b-3118-4077-8509-4c5ed6976f85*
parent: *2681245a-e5c6-43fd-81bf-6e1dd9fb2adf*
children: *49c47513-4f2e-4886-9e79-e32a6b239bf5*

how much power does a human brain consume in watts?

⏮ beginning 1 | ◂ prev 7 || this 8 || next 9 ▸ | end 12 ⏭

Aurora 4omni Munro (ChatGpt) said **:**

gateway: *gpt-4o*
turn ID: *49c47513-4f2e-4886-9e79-e32a6b239bf5*
parent: *aaa2bb9b-3118-4077-8509-4c5ed6976f85*
children: *aaa29253-e0f9-4ee9-88c5-e273e3323c9b*

The human brain consumes approximately **20 watts** of power on average. This power consumption remains relatively constant whether you're resting or engaged in complex cognitive tasks.

To put it in perspective, the brain, which accounts for about 2% of a person's body weight, uses around **20% of the body's total energy consumption**. This translates to about **300–400 kilocalories per day** in terms of energy used by the brain.

In watts:

- **20 watts** is equivalent to powering a small light bulb. This is remarkably efficient considering the complexity and processing power of the brain, which handles memory, decision-making, and countless other functions with such a low energy requirement.

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<< markdown file @ line 517 >>

Conversation 047, Turnpair 9 of 12 ::

@ 2024-08-27 18:42:08 Tue [Δt 0d 00h 05m 29s]

> ***conversation ID:*** *f373746b-2a0a-49a1-95f4-a8d934b65d13 [conversation 047]*
> ***origin ID:*** *f373746b-2a0a-49a1-95f4-a8d934b65d13 [conversation 047]*
> ***created:*** *2024-08-27 11:37:07 Tue*
> ***updated:*** *2025-02-24 17:59:01 Mon*
> ***GPT ID:*** *NA*

▶ *Notespace:*

Wilder Blair Munro (AKUSA) said :

> ***gateway:*** *20Wetbrain*
> ***turn ID:*** *aaa29253-e0f9-4ee9-88c5-e273e3323c9b*
> ***parent:*** *49c47513-4f2e-4886-9e79-e32a6b239bf5*
> ***children:*** *ad6e8e10-9c71-4eca-8937-63a1ad4e6f79*

what is the approximate wight of a human brain?

⏮ beginning 1 | ◂ prev 8 || this 9 || next 10 ▸ | end 12 ⏭

Aurora 4omni Munro (ChatGpt) said :

> ***gateway:*** *gpt-4o*
> ***turn ID:*** *ad6e8e10-9c71-4eca-8937-63a1ad4e6f79*
> ***parent:*** *aaa29253-e0f9-4ee9-88c5-e273e3323c9b*
> ***children:*** *aaa24854-8671-41c1-b231-a617632051ce*

The approximate weight of the human brain is around **1.3 to 1.4 kilograms** (or **about 3 pounds**). This weight can vary slightly between individuals, with some brains weighing slightly less or more depending on factors like age, sex, and overall health.

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Conversation 047, Turnpair 10 of 12 ::

@ 2024-08-27 18:42:47 Tue [Δt 0d 00h 00m 37s]

> ***conversation ID:*** *f373746b-2a0a-49a1-95f4-a8d934b65d13 [conversation 047]*
> ***origin ID:*** *f373746b-2a0a-49a1-95f4-a8d934b65d13 [conversation 047]*
> ***created:*** *2024-08-27 11:37:07 Tue*
> ***updated:*** *2025-02-24 17:59:01 Mon*
> ***GPT ID:*** *NA*

▶ *Notespace:*

Wilder Blair Munro (AKUSA) said :

> ***gateway:*** *20Wetbrain*
> ***turn ID:*** *aaa24854-8671-41c1-b231-a617632051ce*
> ***parent:*** *ad6e8e10-9c71-4eca-8937-63a1ad4e6f79*
> ***children:*** *c2fe9b26-a5fe-47cc-9fa8-f0c7d5d4f6c0, 79049987-e576-4273-a669-6e9fbc0146f2*

what is the weight of the amount of silicon required to train and host a tera-parameter LLM?

⏮ beginning 1 | ◂ prev 9 || this 10 || next 11 ▸ | end 12 ⏭

Aurora 4omni Munro (ChatGpt) said :

> ***gateway:*** *gpt-4o*
> ***turn ID:*** *c2fe9b26-a5fe-47cc-9fa8-f0c7d5d4f6c0*
> ***parent:*** *aaa24854-8671-41c1-b231-a617632051ce*
> ***children:*** *aaa2bca4-6b6a-4ba8-b1ad-01f321742d27*

Estimating the weight of the silicon required to train and host a trillion-parameter (tera-parameter) LLM depends on the hardware used, such as GPUs or TPUs, and how many of these devices are involved in the training and hosting processes.

Breakdown of the Estimation:

1. **Hardware Overview**:
    - Large-scale language models are typically trained on **GPUs** (such as NVIDIA A100, V100) or **TPUs** (Google's Tensor Processing Units).
    - Each GPU/TPU chip is made primarily of silicon and other materials (such as metals and plastic packaging). However, we're focusing on the silicon weight here.

2. **Weight of a Typical GPU/TPU**:
   - The **NVIDIA A100 GPU**, one of the most advanced for AI workloads, weighs approximately **1.5 to 2 kilograms** (the entire unit including casing, cooling system, etc.).
   - The **Google TPU v4 chip** weighs about **2 kilograms** per unit (again, this includes all components, not just silicon).

3. **Silicon Content**:
   - The silicon in a typical chip accounts for a small fraction of the total weight. Silicon wafers are thin, and much of the chip's weight comes from other materials like the packaging, heat sinks, and interconnects.
   - The **silicon die** itself typically weighs around **10–20 grams** per GPU or TPU unit. This is a rough estimate, as it depends on the exact chip design and manufacturing process.

4. **Scale of Infrastructure for Training**:
   - For GPT-3 (175 billion parameters), **10,000 NVIDIA V100 GPUs** were used for training.
   - A **trillion-parameter** model like GPT-4 would likely require a much larger infrastructure. Let's assume **20,000 GPUs/TPUs** for this scale of computation.

5. **Total Silicon Weight for Training**:
   - If each GPU or TPU contains around **10–20 grams of silicon**, and you use **20,000 GPUs/TPUs**:

   $$\text{Total silicon weight} = 20,000\,\text{chips} \times 0.015\,\text{kg}\ (15\ \text{grams}) = 300\,\text{kg of silicon}.$$

6. **Hosting Infrastructure**:
   - Hosting a trillion-parameter LLM also requires many GPUs/TPUs for inference. This infrastructure would likely overlap with training infrastructure to some extent, but the same order of magnitude (thousands to tens of thousands of GPUs/TPUs) would be involved.

Conclusion:

The estimated **weight of silicon** required to train and host a trillion-parameter LLM is around **300 kilograms (660 pounds)**, based on the number of GPUs/TPUs and the amount of silicon in each unit. This is a rough estimate and could vary based on hardware efficiency and specific model infrastructure.

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Aurora 4omni Munro (ChatGpt) said :

> ***gateway:*** *gpt-4o*
> ***turn ID:*** *79049987-e576-4273-a669-6e9fbc0146f2*
> ***parent:*** *aaa24854-8671-41c1-b231-a617632051ce*
> ***children:***
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> ◡○-○ ~~~ ~~~ ~ ~~~ ~ ~~~ ~ ~~~ ~~~ ~ ~~~ ~~~ ~~~ ~ ~~~ ~~~ ~~~ ~ ⟡

Estimating the weight of the amount of silicon required to train and host a tera-parameter large language model (LLM) involves a few assumptions and approximations, given the complexity of modern computing hardware.

Components to Consider:

1. **GPUs/TPUs**: These are the primary hardware components used in training and hosting large LLMs.
2. **Memory and Storage**: High-speed memory (like HBM or GDDR for GPUs) and storage (like SSDs) are also essential, though they contribute less to the total silicon weight.
3. **Supporting Infrastructure**: Includes the interconnects, cooling systems, and power supplies, but these typically involve other materials beyond just silicon.

Estimating the Silicon Content:

1. **Weight of a Single GPU/TPU**:
   - A modern GPU like the NVIDIA A100 or H100, which is commonly used for training large models, typically weighs about 1-2 kg. However, not all of this weight is silicon; much of it is in the metal casing, cooling system, and other components.
   - The actual silicon die in a high-end GPU might weigh **a few grams to tens of grams**. For instance, the silicon die of an A100 might weigh around **20-50 grams**.

2. **Number of GPUs/TPUs**:
   - Training a tera-parameter LLM like GPT-4 might require **thousands of GPUs/TPUs** during training.
   - For hosting and inference, the number of GPUs used depends on the scale of deployment. Large-scale services might use thousands of GPUs or TPUs to handle requests in real-time.

3. **Total Silicon Weight**:
   - Assuming a large-scale deployment involves **10,000 GPUs**, and each GPU contains **50 grams of silicon**, the total silicon weight would be:

$$10,000 \times 50\,\text{grams} = 500,000\,\text{grams} = 500\,\text{kg}$$

Approximate Total Weight:

- The silicon required to train and host a tera-parameter LLM could weigh on the order of **hundreds of kilograms**, potentially around **500 kg** or more, depending on the specific hardware used and the scale of deployment.

Additional Considerations:

- This estimate primarily accounts for the silicon in the processors themselves. The overall infrastructure, including power supplies, cooling systems, and network hardware, would add significantly to the total weight but involve materials other than silicon.
- Memory (e.g., HBM or GDDR) and storage also contain silicon, but their contribution is smaller compared to the GPUs/TPUs.

In summary, the silicon required to train and host a tera-parameter LLM could weigh **hundreds of kilograms**, with a rough estimate being around **500 kg** for the silicon in the GPUs/TPUs alone.

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Conversation 047, Turnpair 11 of 12 ::

@ 2024-08-27 18:46:38 Tue [Δt 0d 00h 03m 26s]

> ***conversation ID:*** *f373746b-2a0a-49a1-95f4-a8d934b65d13 [conversation 047]*
> ***origin ID:*** *f373746b-2a0a-49a1-95f4-a8d934b65d13 [conversation 047]*
> ***created:*** *2024-08-27 11:37:07 Tue*
> ***updated:*** *2025-02-24 17:59:01 Mon*
> ***GPT ID:*** *NA*

▶ *Notespace:*

Wilder Blair Munro (AKUSA) said :

> ***gateway:*** *20Wetbrain*
> ***turn ID:*** *aaa2bca4-6b6a-4ba8-b1ad-01f321742d27*
> ***parent:*** *c2fe9b26-a5fe-47cc-9fa8-f0c7d5d4f6c0*
> ***children:*** *90020b7c-fbfa-43a6-8185-93ff8347bb8c*
>

what's the name for when somebody is stuck in their bed due to medical reasons?

⏮ beginning 1 | ◂ prev 10 || this 11 || ~~next None ▸~~ | end 12 ⏭

Aurora 4omni Munro (ChatGpt) said :

> ***gateway:*** *gpt-4o*
> ***turn ID:*** *90020b7c-fbfa-43a6-8185-93ff8347bb8c*
> ***parent:*** *aaa2bca4-6b6a-4ba8-b1ad-01f321742d27*

> ***children:*** *aaa29ee8-36ba-4bfa-93b7-010d702266ab*
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> ◡○-○ ~~~ ~~~ ~ ~~~ ~ ~~~ ~ ~~~ ~~~ ~ ~~~ ~~~ ~~~ ~ ~~~ ~~~ ~~~ ~ ✧

The term for when someone is stuck in their bed due to medical reasons is **bedridden** or **bed-bound**. This condition occurs when a person is unable to leave their bed for extended periods due to illness, injury, or physical limitations, requiring continuous rest and care.

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Conversation 047, Turnpair 12 of 12 ::

@ 2024-08-27 19:15:04 Tue [Δt 0d 00h 28m 23s]

> ***conversation ID:*** *f373746b-2a0a-49a1-95f4-a8d934b65d13 [conversation 047]*
> ***origin ID:*** *f373746b-2a0a-49a1-95f4-a8d934b65d13 [conversation 047]*
> ***created:*** *2024-08-27 11:37:07 Tue*
> ***updated:*** *2025-02-24 17:59:01 Mon*
> ***GPT ID:*** *NA*

▶ *Notespace:*

Wilder Blair Munro (AKUSA) said :

> ***gateway:*** *20Wetbrain*
> ***turn ID:*** *aaa29ee8-36ba-4bfa-93b7-010d702266ab*
> ***parent:*** *90020b7c-fbfa-43a6-8185-93ff8347bb8c*
> ***children:*** *a5ad0e33-1a70-48a0-94fc-1870f3eee43a*
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what is 10^15*10^9

⏮ beginning 1 | ◂ prev 10 || this 11 || ~~next None ▸~~ | end 12 ⏭

Aurora 4omni Munro (ChatGpt) said :

> ***gateway:*** *gpt-4o*
> ***turn ID:*** *a5ad0e33-1a70-48a0-94fc-1870f3eee43a*
> ***parent:*** *aaa29ee8-36ba-4bfa-93b7-010d702266ab*
> ***children:*** *b84e78c4-aa2e-4032-a4ef-e1d7e701bcdd*
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Perform the calculation

---

result = 10**15 * 8 * 10**9
result

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Aurora 4omni Munro (ChatGpt) said :

gateway: *gpt-4o*
turn ID: *ccb9ba8a-2314-42c5-bcfb-f9e8e8bf7666*
parent: *b84e78c4-aa2e-4032-a4ef-e1d7e701bcdd*
children:

The result of $10^{15} \times 8 \times 10^9$ is **8 x 10^{24}** or **8,000,000,000,000,000,000,000,000**.

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